

1. Brainstorming & Ideation Phase

Project: PocketSmart AI - Smart Budget & Recommendation Assistant

1.1 Introduction

The brainstorming and ideation phase is the starting point of the PocketSmart AI project. In this stage, the project idea is developed by identifying a practical problem faced by users and thinking about how artificial intelligence can provide a useful solution. PocketSmart AI was proposed as a GenAI-powered recommendation assistant that connects budgeting with product and service suggestions.

1.2 Problem Identification

People often need to manage a fixed budget while choosing products or services for different life needs. Examples include decorating a home, organizing an event, or selecting jewelry for a special occasion. The large number of products, platforms, sellers, and price ranges can make the selection process difficult and time-consuming.

The project documentation identifies this challenge as the main problem to solve. A user may know the total amount available to spend, but still need help deciding how to divide that budget among different requirements and which options fit the user's preferences.

1.3 Initial Brainstorming

During brainstorming, the project concept was shaped around three main questions: What does the user need? What is the user's available budget? And which products or services can satisfy the need without exceeding that budget? These questions led to the idea of an intelligent assistant that can understand user inputs and return personalized recommendations.

1.4 Basic Idea

The basic idea of PocketSmart AI is to combine user budget, preferences, contextual requirements, and optional image input with a generative AI recommendation system. The system is intended to make the planning and selection process more convenient and user-friendly.

Page 2 - Idea Development and Objectives

2.1 Development of the Project Idea

The initial idea was expanded from a simple budget calculator into a broader recommendation assistant. Instead of only showing how much money is available, PocketSmart AI is designed to interpret the purpose of the budget and suggest suitable products or services. This makes the system more useful for practical planning.

The concept also uses a cross-platform approach. The project documentation describes recommendations from popular platforms such as Amazon, Flipkart, IKEA, Swiggy, Zomato, and OYO. The purpose is to bring relevant options from different categories into one planning experience.

2.2 Main Objectives

The first objective is to help users stay within a defined budget. Recommendations are intended to consider the amount entered by the user and the requirements associated with the selected planner.

The second objective is personalization. The system takes user preferences and contextual information into account instead of giving the same recommendations to every user.

The third objective is convenience. By combining planning inputs and recommendation generation in one application, users can reduce the effort required to search through many separate platforms.

The fourth objective is to support different types of planning through separate planner modules. The project documentation identifies Home Interior, Party, and Jewelry planning as the major use cases.

2.3 Expected User Experience

The expected experience is simple: the user selects a planner, enters the relevant details, provides a budget, and receives structured recommendations. The interface is intended to use adaptive forms and clear result displays so that users can understand the suggestions easily.

2.4 Role of Generative AI

Gemini 1.5 Flash Pro is described in the project documentation as the AI layer used for understanding budget context, analyzing user inputs, generating structured suggestions, and supporting image analysis for the Jewelry Planner.

Page 3 - Core Ideas and Planner Modules

3.1 Home Interior Planner

The Home Interior Planner was identified as one of the main ideas during the brainstorming stage. Users can enter a home budget and specify requirements for areas such as the living room, kitchen, or bedroom. They can also provide quantities for items such as lights, ceiling fans, and dining tables.

The AI is intended to process these details and generate cost-effective options. The project documentation states that recommendations are balanced across functionality, style, and price so that users can plan their interior needs without overspending.

3.2 Party Planner

The second major idea is the Party Planner. Users provide information such as total budget, guest count, event type, and venue details. The system then considers areas such as catering, decoration, entertainment, and accommodation-related needs when applicable.

The concept is intended to support different event types, including birthdays, corporate events, and weddings. The AI-based approach is designed to organize suggestions according to the available budget rather than leaving the user to perform all the planning manually.

3.3 Jewelry Planner

The third idea is the Jewelry Budget Planner. Users can enter their budget, occasion, and preferred style. An optional outfit image can also be provided so that the AI can consider color coordination and overall aesthetics.

The system is then designed to produce jewelry options that match the selected occasion and style while also remaining aligned with the user's financial plan. This module introduces multimodal interaction because both textual information and images may be used.

3.4 Common Principle Across All Modules

Although the three planners have different purposes, they share the same central principle: collect the user's requirements, understand the budget and context, and generate recommendations that are relevant to the selected use case.

Page 4 - Technology and Recommendation Concept

4.1 Technology Concept Identified During Ideation

The project documentation describes a web-based architecture in which a backend service communicates with the Gemini AI layer and the frontend collects user information. The documented technologies include FastAPI for backend services, HTML/CSS and JavaScript for the frontend, and third-party platforms or APIs for product and service data sourcing.

4.2 Input and Context Handling

A key idea in the brainstorming phase is that recommendations should be based on context rather than on a single keyword. Relevant inputs may include budget, category, preferences, room type, quantity, event type, guest count, occasion, style, and optional image information.

4.3 Recommendation Generation

The recommendation engine is planned to interpret the collected information and generate structured suggestions. The project workflow describes prompt templates for individual planners, budget formatting, category segmentation, and image analysis for multimodal tasks.

4.4 Cross-Platform Concept

Another important idea is to connect recommendations with multiple platforms. The documentation refers to Amazon, Flipkart, IKEA, Swiggy, Zomato, and OYO as examples of sources for product or service suggestions. The intended benefit is to allow users to explore relevant options across different ecosystems from a single application.

4.5 Personalization and User Accounts

The idea also includes user registration, login, session handling, and recommendation history. These features are intended to support personalized interaction and allow users to review previous recommendation queries and results.

4.6 Usability Considerations

The UI concept emphasizes responsive forms, category selectors, dynamic result displays, and card-like recommendation layouts. The purpose is to present product names, details, prices, and other recommendation information in a clear format.

Page 5 - Final Ideation Outcome

5.1 Finalized Project Concept

The brainstorming process resulted in the PocketSmart AI concept: an intelligent, budget-aware assistant for everyday planning and recommendation needs. The system combines structured user inputs with generative AI to create personalized suggestions for home interiors, parties, and jewelry selection.

5.2 Key Features Emerging from Ideation

The final concept includes budget-based planning, personalized recommendations, multiple planning categories, optional image input for jewelry matching, cross-platform product or service sourcing, user authentication, session-based personalization, and recommendation history.

5.3 Expected Benefits

The project is designed to reduce the effort involved in comparing many products and services. It also aims to make budget planning more organized by connecting spending limits with specific recommendations. The use of AI adds the ability to process contextual preferences and generate suggestions dynamically.

5.4 Scope Identified for Further Development

The project workflow identifies later development stages such as Gemini API setup, backend route creation, frontend development, real-world testing, prompt optimization, input validation, and fallback recommendations when AI results are insufficient. These activities follow naturally from the initial brainstorming and provide a path from the idea to a working application.

5.5 Conclusion

The brainstorming and ideation phase established the main direction of PocketSmart AI. The project began with the problem of managing budgets and choosing suitable options across different needs. The solution evolved into a GenAI-powered recommendation assistant that combines budget information, user preferences, contextual details, and optional images. The three main planners - Home Interior, Party, and Jewelry - provide the foundation for the application's functionality.

Overall, the ideation phase provided a clear project concept, defined the major modules, identified the role of AI and supporting technologies, and established the expected user-focused outcome: budget-friendly, personalized, and convenient planning assistance.